JÖNKÖPING UNIVERSITY
School of Engineering

AI-Assisted vs Manual Webpage Layout Design

A feasibility study of performance, experience, and workflow patterns

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Abstract

AI-assisted design tools are increasingly embedded in professional design practice, where they are presented as a means to accelerate workflows and improve efficiency. However, the effects of AI assistance on task performance, perceived workload, and the designer's sense of control remain insufficiently understood, particularly in precision-oriented tasks where accuracy and structural consistency are central requirements.

This study examined how AI-assisted and manual UI design workflows differ in terms of task performance, perceived workload, and workflow strategy in a constrained layout replication task. A controlled between-subjects experiment was conducted with ten design students, five completing the task manually in Figma and five using the AI-assisted Figma Make feature. Performance was assessed using a rubric-based section evaluation, workload was measured using the NASA Task Load Index, and control and confidence were captured through Likert-scale items. Qualitative open-ended responses were analysed to describe workflow patterns across conditions.

The results showed that the AI-assisted condition achieved higher section-level layout scores overall, while the manual condition was associated with stronger perceived control over both the process and the final result. Perceived workload was lower in the AI-assisted condition across most dimensions, though mental demand remained similar across both groups. Qualitative responses revealed distinct workflow strategies: manual participants followed a direct construction approach, while AI-assisted participants followed a generate-and-correct approach. Participants in both conditions expressed a preference for a hybrid workflow combining AI-generated structure with manual refinement.

The findings suggest that AI assistance redistributes effort and control across the design process rather than simply improving outcomes, and that efficiency gains may come at the cost of perceived agency.

Keywords: AI-assisted design, manual design, Figma Make, layout replication, NASA-TLX, perceived workload, perceived control, human-AI interaction, workflow patterns, UI design

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1 Introduction

Artificial Intelligence (AI) has become a recurring component in contemporary design, where it is used to assist designers with tasks such as layout composition, structural organisation, and visual suggestion. AI-based features are now directly ingrained into professional design tools and constitute as a part of routine design practice rather than being positioned as separate experimental systems (Gmeiner et al., 2023). These features are commonly presented as means to work faster and more efficiently by automating or accelerating time-consuming design activities and allowing for rapid generation and exploration of alternatives, particularly at the early stages of the design process (Yu et al., 2025). In a similar manner, integrating AI into the design process changes not only the final result but also the way designers work.

Simultaneously, the use of AI in design introduces new forms of work related to interacting with and managing system-generated outputs. While AI can facilitate the initiation of tasks, research indicates that it can also present challenges related to supervision, verification, and correction (Amershi et al., 2019). Designers remain responsible for ensuring accuracy and relevance, particularly in the later stages of work where higher levels of precision are required. Consequently, rather than eliminating labour, AI may redistribute effort across different phases of the design process (Salma et al., 2025).

These demands for supervision, verification, correction, and precision are especially relevant in interface design tasks, where the outcomes depend on maintaining consistent visual hierarchy, precise spacing, and clearly defined structure. Worku and Kirwa (2025) describe AI-supported design as a guided and iterative process in which designers continuously evaluate, refine, and adapt AI-generated outputs rather than passively accepting them. Their findings suggest that while AI can accelerate early-stage workflow activities and support repetitive tasks, designers still rely heavily on human judgement when accuracy, contextual understanding, and quality control become central requirements. This supports the idea that AI may not necessarily remove effort from the design process but instead redistribute it towards supervision, refinement, and decision-making.

Fan et al. (2022) further argue that the structure of human-AI collaboration can influence how users engage with tasks and respond to AI-generated suggestions throughout their workflow. Their findings demonstrate that interaction design and explanatory support may shape user behaviour, engagement, and perceptions of AI-supported systems during task completion.

1.1 Problem statement

As AI becomes an active component within human-technology systems, its role extends beyond automation to shaping how responsibility, control, and effort are distributed during the work process. Research in human-centred AI emphasises that AI systems influence responsibility, control, and workload by redistributing decision-making and monitoring effort between humans and machines (Shneiderman, 2020). As a result, the impact of AI on design work cannot be assessed solely by output quality or speed, but must also consider how AI changes task distribution and affects workload, perceived control, and workflow management.

Studies in human-AI interaction show that AI-assisted systems often operate under uncertainty and produce variable outcomes, requiring users to supervise, verify, and correct system outputs (Amershi et al., 2019; Shneiderman, 2020). These demands can affect perceived control, trust, and workload, particularly when AI behaviour is not fully transparent or predictable (Amershi et al., 2019). Rather than eliminating human effort, AI systems can introduce additional cognitive and operational work related to error handling, quality assurance, and decision validation.

Research on technologically mediated work further demonstrates that automation tends to redistribute labour rather than remove it (Kittur et al., 2013). Studies on crowd work, for example, show how technological systems shift effort toward coordination, verification, and quality control, creating forms of “hidden work” that remain necessary for achieving reliable outcomes (Kittur et al., 2013). This pattern extends beyond crowdsourcing contexts and is relevant to professional domains in which humans collaborate with automated systems.

In the field of design, these dynamics are reflected in human-AI co-creative systems, where AI generates suggestions while designers retain responsibility for evaluating, refining, and finalising outcomes. Although such systems can support early task progress, they may also require increased effort in later stages due to refinement, correction, and verification, particularly in tasks that demand high accuracy and consistency (Liu, 2025; Salma et al., 2025). This suggests that AI assistance may change where effort occurs within a workflow rather than simply reducing overall effort.

AI-assisted design research has primarily examined open-ended activities such as ideation, creativity, and general usability (Fan et al., 2022; Liu, 2025). In contrast, tasks such as precise replication of an existing interface represent a distinct context in which accuracy, consistency, and adherence to predefined structures are central.

This gap presents a significant and scientifically pertinent issue concerning how AI-assisted workflows function under conditions of high precision. In particular, it remains unclear how AI support influences performance, perceived workload, and the management of the design process when accuracy is prioritised over exploration. By addressing this issue, it can broaden a more grounded understanding of the trade-offs associated with AI assistance, along with serving as a starting point for investigating how different workflows shape both outcomes and user experience in controlled design tasks.

1.2 Purpose and research questions

Drawing on the problem statement, it is evident that AI is increasingly integrated into design workflows and is discussed as a driver of automation and efficiency in graphic design processes, while also introducing challenges (e.g., limitations in understanding aesthetics/emotion and risks such as homogenization) (Li et al., 2024).

Further, research on human-AI collaboration suggests that AI-supported systems are shaped not only by the quality of generated outputs but also by how users interact with, interpret, and respond to AI-generated suggestions throughout the workflow (Fan et al., 2022). Factors such as explanatory support, interaction structure, and user engagement may therefore influence how designers work with AI-assisted tools during task completion.

Consequently, the purpose of this study is to examine how AI-assisted and manual UI design workflows differ in terms of task performance and user experience in a constrained layout replication task.

Prior work often frames AI as improving efficiency in design workflows while also emphasising limitations that may affect quality and consistency (Li et al., 2024). This motivates a controlled comparison that considers both performance outcomes and the designer's experience of the work process.

Based on this motivation, the research questions are:

[1] To what extent does AI assistance affect task performance compared with a manual workflow?

[2] What impact does AI assistance have on perceived workload compared with manual work?

[3] How do designers' workflow strategies and decision-making differ between AI-assisted and manual work?

1.3 Scope and limitations

This study examines AI-assisted versus manual interface layout work, focusing on performance, participants' experience, and workflow differences when completing the same design goal. The study uses a replication task to provide a shared target outcome and to reduce task-driven variation from multiple equally valid design solutions, which supports clearer comparisons between conditions in a controlled experimental setup (Hornbæk, 2011). The scope therefore centres on performance, participants' experience, and workflow differences when completing the same layout goal with and without AI assistance.

The study deliberately excludes learning effects, skill development, and transfer to future tasks because it uses a single-session design and focuses on comparing the two workflows within one task context. It also excludes creative outcomes, originality, aesthetic preference, and end-user outcomes such as usability, engagement, or conversion, because the task prioritises matching a provided reference layout rather than producing a "better" or more novel design solution.

The AI condition is limited to the Figma Make feature, so the findings should be interpreted as tool- and context-specific rather than representative of AI-assisted design tools in general. With five participants per condition, the results are exploratory and intended to describe patterns and effect directions rather than support broad generalisation.

1.4 Disposition

Chapter 2 presents the theoretical framework that motivates the study and defines the key concepts used in the analysis. Chapter 3 describes the method and implementation, including how data are collected, how the data are analysed, how validity and reliability are addressed, and which considerations guide the study design. Chapter 4 reports the results for the two conditions in terms of performance, experience, and observed workflow patterns. Chapter 5 discusses the results and provides a method discussion that reflects on the study design, limitations, and interpretation of findings. Chapter 6 concludes the thesis by summarizing the main conclusions and outlining directions for further research.

2 Theoretical framework

2.1 Design Tools and Iterative Design Practice

Design work in professional digital contexts is fundamentally shaped by the tools through which it is carried out. Research on design practice emphasises that tools are not neutral instruments but actively mediate how designers understand problems, explore alternatives, and transform intentions into concrete outcomes (Stolterman, 2008). In interaction design, work is typically situated within constrained and complex situations, where progress depends on designers' ability to use tools to manage uncertainty, make judgements, and arrive at satisfactory solutions within limited time and resources (Stolterman, 2008). From this perspective, design is understood as a goal-orientated and iterative activity in which decision-making is closely intertwined with tool use.

Design tools, therefore, function not only as means of execution but also as instruments that shape how design situations are perceived and evaluated. Dalsgaard (2017) argues that tools play a central role in design inquiry by supporting the externalisation of ideas, the framing of problems, and the assessment of potential solutions through ongoing interaction. As a result, changes in tool capabilities can alter how effort, attention, and responsibility are distributed across the design process. This understanding provides a theoretical foundation for examining how new forms of tool support may reshape design work, rather than simply making existing practices faster or more efficient.

Human-Computer Interaction (HCI) is an interdisciplinary field of research that focuses on understanding how people interact with computer systems along with designing interactive technologies that support effective, efficient, and satisfying use (Wang et al., 2021). Within HCI research, user interface design has been described as a process that unfolds through ongoing movement between different levels of abstraction rather than through a fixed sequence of steps (Granlund et al., 2001). Designers work back and forth between understanding tasks, organising structure and navigation, and producing concrete interface elements, while decisions made at one level frequently require reassessment at others. This understanding is consistent with research on user interface software tools, which characterises interface design as a reflective activity focused on maintaining coherence across goals, structure, and interface form through ongoing evaluation (Myers et al., 2000).

This perspective emphasises iteration and refinement as inherent aspects of interface design work. Rather than being defined upfront, interface solutions are gradually shaped through repeated cycles of construction, evaluation, and adjustment (Granlund

et al., 2001). Research on iterative user-interface design further highlights that interface elements are highly interdependent, such that even small changes necessitate verification across multiple parts of the design (Nielsen, 1993). As a result, correction and refinement become integral parts of the workflow rather than final stage activities.

Balasubramanian et al. (1998) present a similar understanding, framing UI design as a non-linear problem-solving activity. They argue that designers repeatedly shift between abstract reasoning about the problem and concrete representations of possible solutions as their understanding develops. This view aligns with Cross's characterisation of design as a process in which problem framing and solution development evolve together through iterative reasoning across different levels of abstraction (Cross, 2018).

2.2 Human-AI Co-Creation and Collaborative Design

Human-AI co-creativity research frames AI not as a fully autonomous creator but as a collaborative partner that participates in the design process through iterative exchange with the human designer. Recent HCI research characterises co-creative systems as mixed-initiative environments in which both the human and the AI can propose, revise, and evaluate design ideas across multiple stages of the workflow (Kadenhe et al., 2025). In these settings, AI-generated outputs serve as material for reflection rather than final solutions, requiring designers to interpret, select, modify, and sometimes reject system suggestions in relation to their goals (Liu, 2025). This shifts creative work away from direct construction and toward decision-making, evaluation, and refinement, which further aligns with views of design as an iterative and non-linear process rather than a sequence of fixed steps (Kadenhe et al., 2025).

Rezwana and Maher (2022) further emphasise that the design of interaction itself plays an important role in shaping human-AI collaboration. Their study found that AI systems capable of communicating back to users through feedback, conversational responses, and guidance were more frequently perceived as collaborative and engaging compared to systems limited to one-way interaction. Participants described communicative AI systems as feeling more attentive, supportive, and involved throughout the creative process, while limited interaction made AI feel closer to a passive image-generation tool (Rezwana & Maher, 2022). These findings suggest that collaborating with AI is influenced by both the quality of generated outputs and how the system participates in the workflow and responds during task completion.

Research also indicates that this collaborative dynamic can produce mixed effects on designers' experience of control and authorship. While AI support can reduce cognitive load and help designers overcome blank-canvas situations by providing rapid alternatives, it can also introduce uncertainty about ownership and responsibility

for design decisions (Liu, 2025). Studies of human-AI co-creation report that designers may feel they are reacting to system-generated suggestions rather than actively steering the process, particularly when AI initiative becomes difficult to predict or interpret (Kadenhe et al., 2025). As a result, designers often invest additional effort in verification, correction, and alignment with predefined requirements to reassert control over the outcome (Liu, 2025). This perspective supports examining AI-assisted design not only in terms of efficiency gains but also in terms of how creative agency, confidence, and perceived control are redistributed across the design process.

2.3 Generative AI and Changing Design Workflows

Advances in generative artificial intelligence (GenAI) have made it possible for design tools to incorporate models that can generate text, images, and interface structures based on learned patterns from large datasets, which enables parts of the design process to be computationally produced rather than manually constructed (Bommasani et al., 2021). Generative features built into design tools can change more than speed. They may also alter how work is distributed across the process. Some steps are carried out by the system or suggested to the designer, while the designer's role shifts toward setting intent, choosing among options, and checking the final result. For this reason, AI should not be treated as a neutral "speed button" in this study. It is treated as a form of support that can shape decision-making and the way designers experience competence and control while working.

AI-augmented design support serves as a central perspective in this study. Fan et al. (2022) describe human-AI collaboration as a process in which users continuously negotiate between their own judgement and AI-generated suggestions during task completion. Rather than functioning as a passive tool, AI becomes an active part of the workflow that can influence how users analyse information, prioritise actions, and approach decision-making throughout a task. This helps explain why AI support may feel beneficial in some situations, while in others it may introduce hesitation, additional checking, or changes in working strategy despite potential gains in speed.

2.3.1 Verification, Agency, and the Cost of AI Assistance

It is also important to understand why the effects of AI support can be mixed. Work on AI in graphic design points out that such tools can speed up visual production and increase the number of possible solutions, but they can also add new kinds of work. Designers may spend more time selecting, judging fit for the goal, correcting errors, and aligning the output with strict requirements (Li et al., 2024). In other words, efficiency can improve, while the "cost of verification and refinement" can increase.

A related issue concerns agency and authorship. Even when AI helps designers get started quickly or produces strong drafts, it can change the feeling of "I am steering this" versus "I am reacting to what the system gave me." This can show up as uncertainty about ownership and reduced confidence in decisions. Studies of human-AI co-creation report that AI support can lower perceived cognitive load while still raising questions about blurred authorship and control when the boundary between human contribution and system contribution becomes unclear (Liu, 2025). This is one reason why this thesis looks beyond outcome measures and also considers workload, confidence, and perceived control.

2.4 Theoretical Positioning

Overall, this theoretical framework positions design work as an iterative and tool-mediated activity in which decision-making, evaluation, and refinement are closely intertwined. It highlights that both traditional design tools and AI-augmented systems actively shape how designers distribute effort, make decisions, and maintain control throughout the process (Stolterman, 2008; Dalsgaard, 2017). Within interface design, the need for precision and consistency further emphasises the importance of verification and iterative refinement (Nielsen, 1993; Granlund et al., 2001).

From a human-AI co-creation perspective, AI is understood as a collaborative partner that contributes suggestions while requiring continuous human evaluation, selection, and adjustment (Liu, 2025; Kadenhe et al., 2025). This introduces potential trade-offs between efficiency and control, as AI may reduce effort in some stages while increasing demands for supervision, correction, and decision validation in others (Li et al., 2024).

3 Method and implementation

This chapter outlines the methodological approach used to address the research questions and to generate empirical insights into AI-assisted and manual design workflows. In design and human-computer interaction research, methodological choices are closely tied to how phenomena are isolated, observed, and interpreted under controlled or real-world conditions (Cairns, 2014). The choice of the research design therefore determines how the study captures both measurable outcomes and the process underlying them.

To examine differences between AI-assisted and manual workflows in a structured and comparable manner, the study used a controlled between-subjects design and combined quantitative and qualitative methods. Experimental approaches are commonly used to examine cause-and-effect relationships under controlled conditions, allowing for systematic comparison between conditions and the isolation of specific variables (Cairns, 2014; Still, 2011). This technique is particularly relevant in the present study, where the aim is to compare two distinct workflow conditions while minimising external influences that could affect performance.

At the same time, the study incorporates a mixed-methods approach in order to capture not only measurable performance outcomes but also participants' experiences of the task. Mixed-methods research enables the integration of quantitative and qualitative data, supporting a more comprehensive understanding of complex circumstances that cannot be fully explained through a single method alone (Johnson et al., 2007). This approach is important in the context of AI-assisted design, where both objective performance and subjective experience, such as perceived control and workload, are central to understanding the effects of AI support.

A between-subjects design was adopted to structure the comparison between conditions. In this setup, each participant is exposed to only one workflow, which limits the influence of learning and carryover effects that may arise when the same individual completes multiple conditions (Charness et al., 2012). This separation supports a clearer comparison between manual and AI-assisted work, as task strategies and performance are not shaped by prior exposure to an alternative workflow. Differences between conditions can therefore be interpreted as outcomes of the experimental setup rather than effects of familiarity or adaptation.

The methodological approach combines controlled experimental comparison with both quantitative and qualitative data collection to examine different aspects of the design process. Performance measures capture observable outcomes such as time and task completion, while self-reported measures and qualitative reflections provide insight into how participants experience and navigate the task.

3.1 Data collection

This section describes how the study collected the data needed to answer the research questions. Data collection combined task performance measures, including completion time and submitted layout, with a post-task questionnaire and short qualitative reflections in order to capture performance, experiences, and workflow context.

3.1.1 Participants and recruitment

Participants were design students recruited through university channels and peer networks. The study required participants who could already use Figma at a basic level; complete novices would confound the results with first-time tool learning, while highly experienced designers could reduce observable differences due to ceiling effects. A student sample therefore provided a feasible middle ground for comparing the two workflows.

The study included 10 participants in total, with 5 participants per condition. The sample size was determined by recruitment feasibility within the thesis timeframe, and the study is treated as exploratory. Similar sample sizes are common in small HCI studies, but they limit generalisability and statistical power (Caine, 2016).

Before the study, participants completed a short pre-task questionnaire designed to collect demographic and background information relevant to the experiment **(Appendix A)**. The questionnaire included questions regarding age range, gender, design background, previous experience with Figma, familiarity with specific Figma features, and prior exposure to AI-based design tools. Participants were also asked to self-assess their confidence and level of experience using both Figma and AI-supported design tools. This information was used to provide contextual understanding of participants' prior familiarity with the types of tools involved in the study and to help ensure that participants possessed at least basic experience relevant to the experimental task.

Background information from the questionnaire was used to describe the sample and to support balanced assignment across conditions. Participants were assigned to one of the two conditions using block randomisation based on self-reported Figma experience to keep group sizes balanced and experience levels as comparable as possible. Any deviations due to scheduling constraints were recorded.

3.1.2 Materials and tools

The task was performed in Figma. A single desktop webpage reference layout was provided as a static image, and a shared starter file was used to standardise setup

across participants **(Appendix C)**. In the manual condition, participants used standard Figma tools only. In the AI-assisted condition, participants used Figma Make as part of the replication workflow.

A clear instruction sheet described the goal of matching the reference layout and explicitly discouraged redesign decisions. Participants received the same constraints, the same reference, content and the same definition of what counted as completion.

3.1.3 Task procedure

The study followed a standard procedure. Participants first received a short briefing describing the goal and constraints. They then completed the task under a fixed time limit of 20 minutes. The time limit standardised time pressure across participants and kept the session duration reasonable. Because time pressure can influence the speed-accuracy trade-off, the study assessed both time outcomes and layout fidelity to interpret performance under the same constraint (Heitz, 2014).

During the session, the study collected performance data and recorded screen activity to support later process observation. Performance data consisted of time-to-completion and the produced layout, which was later assessed using the predefined fidelity rubric.

After completing the task, participants completed the NASA Task Load Index (NASA-TLX) to report perceived workload. In addition, they responded to short Likert-scale items on confidence and perceived control and answered open-ended questions about their workflow and task experience **(Appendix B)**.

Screen activity was recorded for process observation. The recordings were used only for research purposes and were stored in anonymised form.

3.1.4 Measures recorded during collection

Time was recorded as total task duration from the start of work to the participant's declared completion. If a participant did not finish within the 20-minute time limit, the session was recorded as a time-out and the partial outcome was still saved for fidelity scoring.

Layout fidelity was assessed using a predefined rubric applied consistently across participants **(Appendix E)**. The rubric focused on structural alignment with the reference, key spacing relationships, and recognisable visual hierarchy.

Experience was captured using self-reports collected immediately after the task. Workload was measured using NASA-TLX (Hart & Staveland, 1988). Confidence and perceived control were measured with Likert-style items **(Appendix B)**.

Qualitative reflections were collected via open-ended questions. These responses were used to provide context for the quantitative outcomes and to describe workflow and decision-making patterns across conditions.

3.2 Data analysis

Quantitative analysis focused on comparing the two conditions across performance and experience measures. Task time was operationalised as time-to-completion in minutes. Completion within the 20-minute time limit was treated as a separate outcome and reported as a completion rate per condition. Participants who did not complete the task within the time limit were retained as time-outs. For comparison purposes, their time was recorded as 20 minutes, while their partial outputs were included in the fidelity assessment to preserve information about performance under constraint.

Layout fidelity was assessed using the predefined rubric **(Appendix E)**, producing a total score for each submission. These scores were compared between conditions to examine differences in structural accuracy. Individual rubric components were also analysed to identify recurring patterns of mismatches across layouts.

Perceived workload was measured using NASA-TLX, with an overall workload score calculated as the unweighted mean of the six dimensions (Hart & Staveland, 1988). The individual dimensions were additionally examined to capture variation in different aspects of workload. Confidence and perceived control were analysed as ordinal measures based on Likert-scale responses **(Appendix B)**.

Quantitative analysis focused on descriptive comparisons between the two conditions across performance and experience measures. Task time, completion rate, layout fidelity, and perceived workload were summarised using central tendencies and distributional characteristics to provide an overview of the differences between conditions.

Visualisations were used to support the analysis, including bar charts and summary plots that illustrate the differences in performance and workload between the manual and AI-assisted workflows. These representations allow patterns in the data to be communicated clearly without relying on inferential statistical testing.

Given the small sample size and exploratory nature of the study, the analysis emphasises observed trends and differences rather than statistical significance. The

results are therefore interpreted as indicative of how the two workflows compare, providing a basis for further investigation rather than definitive conclusions.

Qualitative responses were analysed using a thematic analysis (Braun & Clarke, 2006). The participant's answers were initially read in their entirety in order to discern recurring patterns in how they described their workflow and decision-making, how they reviewed their work, and how they corrected the design. Subsequent segments of the responses were then labelled with short descriptive tags to facilitate the grouping of similar observations. These groups were then compared and categorised based on common issues among the participants. These issues were used to describe differences and similarities between the manual and AI-assisted conditions, supported by representative examples from the data.

3.3 Validity and reliability

To support validity and reliability, the study used a controlled task setup and standardised procedures across participants in order to reduce variation unrelated to the experimental conditions (Cairns, 2014). Both groups completed the same layout replication task using identical reference material, instructions, and a fixed 20-minute time limit. The time limit was introduced to standardise time pressure and reduce open-ended polishing during the task. As Heitz (2014) notes, time pressure can influence the speed-accuracy trade-off, meaning that faster performance may occur at the expense of accuracy. For this reason, the study assessed both time outcomes and rubric-based fidelity, while non-completion within the time limit was treated as a meaningful outcome rather than excluded from analysis.

3.3.1 Internal validity

A between-subjects design was used to reduce learning and carryover effects that could occur if the same participants completed both conditions using the same reference layout (Charness et al., 2012). Participants were assigned to conditions using randomisation and balancing procedures based on self-reported Figma experience in order to keep group sizes and experience levels as comparable as possible. Despite these precautions, some individual differences may still remain and should therefore be considered when interpreting between-group differences (Cairns, 2014).

3.3.2 Construct validity

The task was defined as high-fidelity layout replication rather than redesign in order to ensure that outcomes reflected performance under constrained conditions rather than open-ended creative choices (Hornbæk, 2011). The selected measures were aligned with the aims of the study: time-to-completion and completion within the

20-minute limit were used to capture efficiency, while the rubric-based fidelity score assessed accuracy relative to the reference layout. Subjective workload was measured using NASA-TLX, a widely used workload assessment tool designed to evaluate perceived demands during task performance (Hart & Staveland, 1988). Additional Likert-scale items were used to capture participants' perceived confidence and sense of control **(Appendix B)** while working with and without AI support.

3.3.3 Reliability

Data collection was standardised through the use of a shared starter file, fixed instructions, and identical task constraints across both conditions. Fidelity assessment was based on a predefined rubric **(Appendix E)** that was applied consistently across all submissions in order to support scoring reliability (Hornbæk, 2011). To reduce judgement bias, the rubric criteria were defined in advance and applied using the same evaluation procedure for each participant. Where feasible, a subset of submissions could be independently reviewed to help identify inconsistencies and improve scoring consistency.

3.3.4 External validity

The findings of this study are expected to generalise primarily to similar contexts involving design students performing constrained webpage layout replication tasks in Figma under time pressure (Cairns, 2014). Generalisation to other forms of design work, professional designers, different webpage structures, or other AI-supported design tools is more limited, particularly because workflows and human-AI interaction patterns may vary depending on the specific task and tool environment (Kadenhe et al., 2025).

3.4 Considerations

Participation was voluntary and based on informed consent, and participants could withdraw at any time without consequences. Only background information relevant to the study was collected.

Screen recordings were used to support process observation. All recordings and survey data were stored in anonymised form, used only for research purposes, and accessed only by the researchers. Results were reported in aggregated form.

To ensure fairness and comparability, both groups received the same reference layout, the same starter file, the same instructions, and the same 20-minute time limit. The only intended difference between conditions was access to Figma Make. The study did not evaluate end-user outcomes such as usability or user preference, and conclusions were limited to this controlled tool-and-task setting.

4 Results

This chapter presents the findings in relation to the study's research questions. In line with the analytical structure described in the method chapter, the results are presented in three parts: performance, experience, and workflow patterns. The findings are reported at the condition level, comparing the manual and AI-assisted groups **(Appendix C)**. Questionnaire responses and design evaluations are presented as anonymised group-level data and are therefore not linked at the individual participant level.

4.1 Performance

Performance was examined through completion within the fixed 20-minute time limit and through rubric-based section evaluation of the submitted layouts. Based on the post-task questionnaire data, all participants were recorded as not having completed the task within the time limit. This applied to both conditions, resulting in five recorded time-outs in the manual condition and five in the AI-assisted condition.

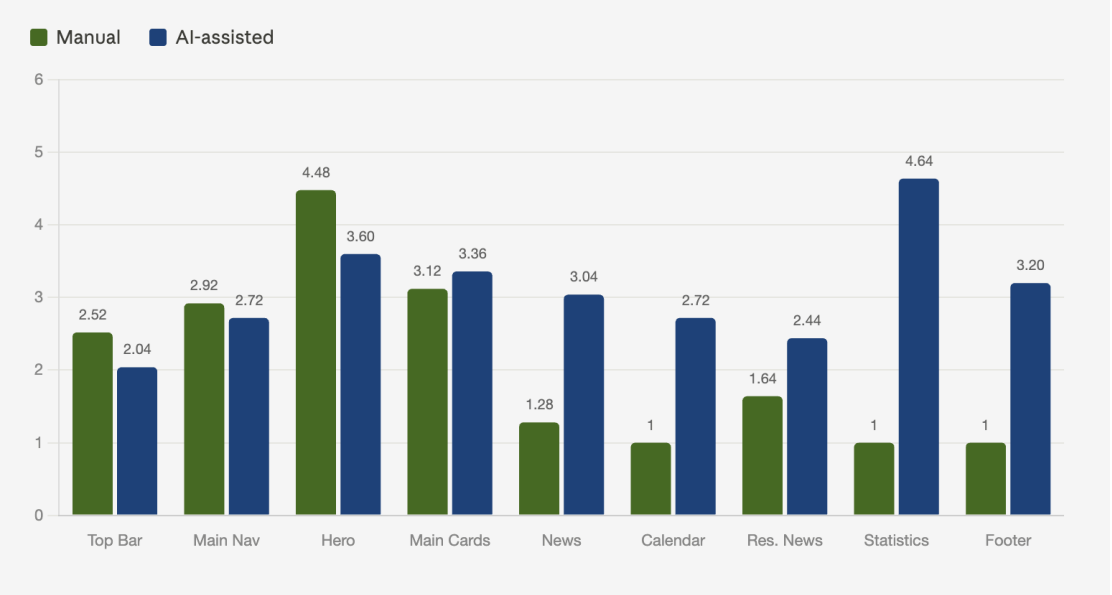


Figure 1. Mean rubric-based section scores for the manual and AI-assisted conditions across nine webpage sections (scale 1–5).

To complement the completion status, the layouts were evaluated section by section using the predefined rubric **(Appendix E)**. Across all evaluated section rows, the manual condition had an average score of 2.11, whereas the AI-assisted condition had an average score of 3.08. Across the five submissions in each condition, the mean section score ranged from 1.73 to 2.69 in the manual condition and from 2.53 to 3.80 in the AI-assisted condition.

The section-based comparison showed different performance patterns across the webpage. In the manual condition, the strongest results appeared in the Hero Section, followed by Main Cards and Main Navigation, while several later sections received considerably lower scores. Since participants generally progressed through the layout from top to bottom under the fixed time constraint, lower scores in later sections likely also reflect reduced available time for completion and refinement. In contrast, the AI-assisted condition showed stronger and more consistent performance across later sections of the webpage, particularly in the Statistics Section and Footer.

As illustrated in Figure 1, the AI-assisted condition achieved higher scores across most middle and later webpage sections, whereas the manual condition performed more strongly in the early navigation-related sections at the top of the page.

Overall, the results indicated that both conditions were constrained by the 20-minute limit, but the AI-assisted condition achieved higher rubric-based section scores across the layout as a whole.

4.2 Experience

Experience was analysed through two components: perceived workload and self-reported control and confidence. Workload was measured using NASA-TLX, while control over the process, control over the final result, and confidence in the submitted result were measured through Likert-scale items **(Appendix B)**.

The descriptive comparison showed a higher overall workload in the manual condition than in the AI-assisted condition. The mean NASA-TLX score in the manual condition was 56.50, compared with 41.17 in the AI-assisted condition. The median values followed the same pattern, with 55.00 in the manual condition and 38.33 in the AI-assisted condition.

Across the individual NASA-TLX dimensions, mental demand was similar across the two conditions, with a mean of 44.00 in the manual condition and 45.00 in the AI-assisted condition. Larger differences appeared in physical demand, effort, and frustration. The manual condition showed higher mean values for physical demand (31.00 vs. 7.00), effort (75.00 vs. 46.00), and frustration (56.00 vs. 43.00). Temporal demand was high in both groups, but remained higher in the manual condition (86.00) than in the AI-assisted condition (65.00). Self-rated performance was slightly higher in the AI-assisted condition (59.00) than in the manual condition (53.00).

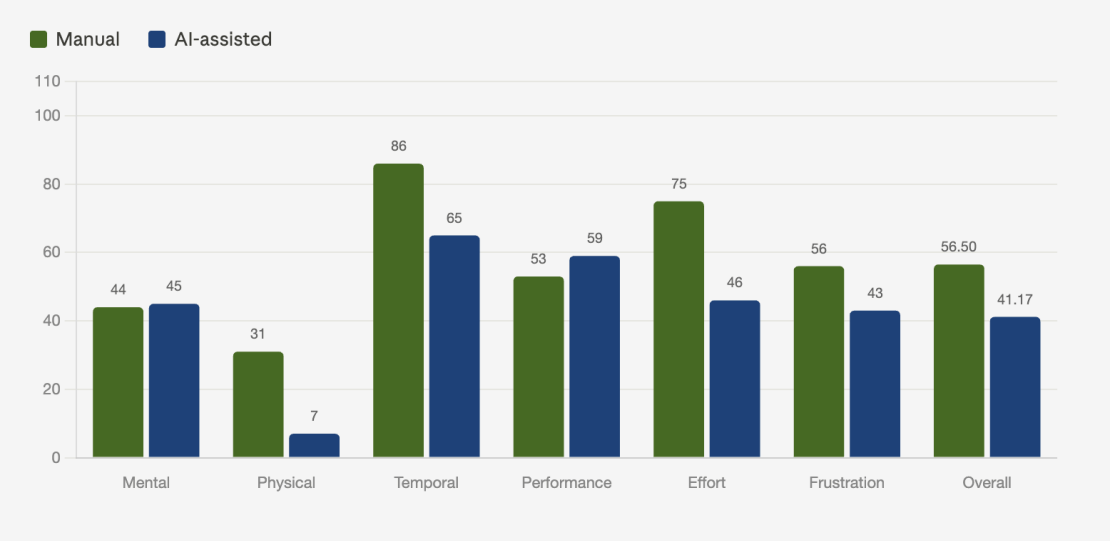


Figure 2. Mean NASA-TLX scores for the manual and AI-assisted conditions across six workload dimensions and overall (scale 0–100).

The Likert-scale responses showed the opposite pattern. Participants in the manual condition reported a higher sense of control and confidence than participants in the AI-assisted condition. The mean overall Likert score was 6.07 in the manual condition and 4.07 in the AI-assisted condition. In the manual condition, the mean rating for control over the process was 6.20, compared with 4.20 in the AI-assisted condition. The same pattern appeared for control over the final result, where the mean was 6.60 in the manual condition and 3.60 in the AI-assisted condition. Confidence was also higher in the manual condition, with a mean of 5.40, compared with 4.40 in the AI-assisted condition.

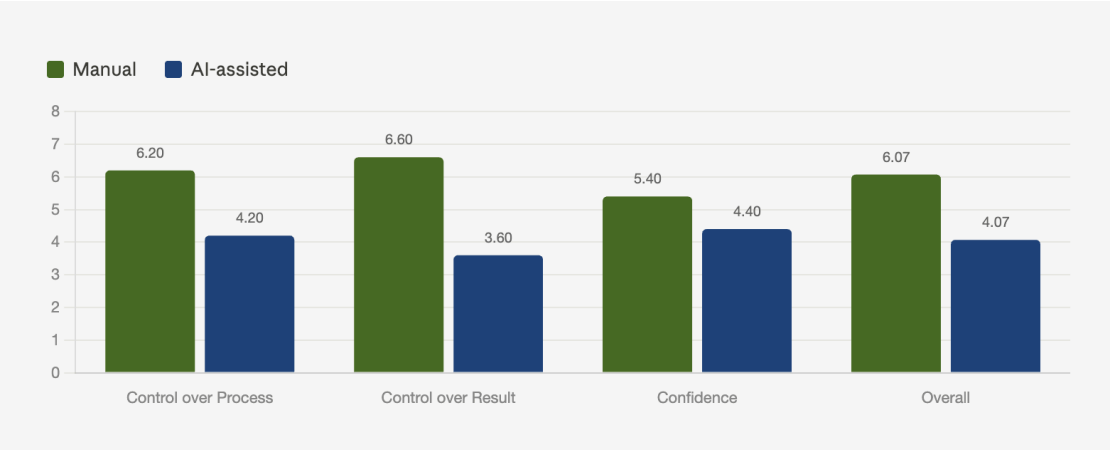


Figure 3. Mean Likert-scale scores for the manual and AI-assisted conditions across three items and overall (scale 1–7).

Taken together, the questionnaire results indicated that the AI-assisted condition was associated with lower perceived workload, while the manual condition was associated with higher perceived control and confidence.

4.3 Workflow

The qualitative responses revealed clear differences in how participants approached the task in the two conditions.

In the manual condition, participants commonly described a direct construction strategy. They typically began by setting up the frame, mapping out the main layout structure, and then placing images, colours, and text. Several participants described working from top to bottom and continuously comparing their output with the reference image. The most frequently reported difficulties in the manual condition concerned spacing, image sizing and cropping, alignment, and the time required to refine smaller visual details.

In the AI-assisted condition, participants more often described a generate-and-correct workflow. They typically began by uploading the reference image, prompting Figma Make to recreate the page, and then correcting sections iteratively to bring the output closer to the reference. A recurring issue in this condition was difficulty in expressing the intended changes clearly enough for the AI system to understand them. Participants also reported repeated prompting, waiting for outputs, and situations where correcting one issue introduced new problems elsewhere in the layout.

Across both conditions, many participants expressed a preference for a hybrid workflow. Automation was most often described as useful for producing an initial draft, overall structure, or repetitive layout tasks. Manual work was preferred for more detailed adjustments, especially spacing, typography, colour, alignment, and final refinement.

Overall, the workflow results suggest that the manual condition was characterised by direct construction and continuous visual comparison, whereas the AI-assisted condition was characterised by prompting, generation, and iterative correction.

5 Discussion

This chapter discusses the results in relation to the purpose, research questions, and theoretical framework. The first part interprets the findings across performance, experience, and workflow. The second part reflects on the strengths and limitations of the study design.

5.1 Result discussion

The purpose of this study was to examine how AI-assisted and manual UI design workflows differ in terms of task performance and user experience in a constrained layout replication task. The results do not point to one workflow being simply better than the other. Instead, they reveal a consistent trade-off between output quality and perceived control that runs across all three aspects of the comparison.

Research question 1 asked to what extent AI assistance affects task performance compared with a manual workflow. The AI-assisted condition produced higher rubric-based section scores overall (3.08 vs. 2.11), with stronger results in sections that appeared later in the page, such as the Statistics Section, Calendar Section, and Footer. At the same time, all participants in both conditions were recorded as time-outs, which means AI assistance did not translate into task completion. This pattern suggests that AI helped participants progress further into the layout within the fixed time, improving section-level output quality, but did not resolve the underlying time pressure of the task. This is consistent with Li et al.'s (2024) observation that AI tools can accelerate certain phases of design work while introducing additional demands elsewhere, such as verification and correction, that prevent straightforward efficiency gains.

One finding worth noting is that the manual condition outperformed the AI-assisted condition in sections that appeared earlier in the page, including Top Bar, Main Navigation, and Hero Section. This may reflect the generate-and-correct workflow described by AI-assisted participants, in which the initial AI output for these sections required repeated correction and refinement before participants could move forward. Rather than saving time at the start, AI assistance may have concentrated effort at the correction stage, which is consistent with what Kadenhe et al. (2025) describe as the ongoing evaluation and adjustment work that characterises human-AI co-creative environments.

Research question 2 asked what impact AI assistance has on perceived workload compared with manual work. The AI-assisted condition was associated with a lower overall NASA-TLX score (41.17 vs. 56.50), with particularly large differences in physical demand, effort, and frustration. However, the Likert-scale responses showed

the opposite direction: manual participants reported higher control over the process, control over the final result, and confidence in the submitted layout. This combination of lower workload alongside lower perceived control is an important finding because it shows that feeling less burdened does not necessarily translate into feeling more capable or in charge.

One unexpected dimension of the workload comparison was mental demand, which was nearly identical across conditions (44.00 manual vs. 45.00 AI-assisted). If AI assistance reduced the cognitive complexity of the task, a lower mental demand score in the AI-assisted condition might be expected. The absence of this difference may suggest that working with AI introduced its own cognitive demands, such as formulating effective prompts, evaluating generated outputs, and deciding which corrections to prioritise, that replaced rather than reduced mental effort. This interpretation aligns with Fan et al.'s (2022) description of human-AI collaboration as a continuous negotiation between the user's own judgement and AI-generated suggestions, which may sustain cognitive engagement even when physical effort and frustration are reduced.

The control and confidence gap is closely related to what Stolterman (2008) describes as the mediating role of tools in design work. When designers construct a layout manually, the tool responds directly and predictably to each action, which supports a clear sense of what is happening at every step and why. In the AI-assisted condition, this direct relationship between action and outcome was replaced by a more indirect process in which participants submitted prompts and received generated outputs that they then had to interpret and adjust. This shift may explain why AI-assisted participants felt less in control despite experiencing less overall effort, a tension that Liu (2025) also identifies in human-AI co-creation contexts where designers report uncertainty about ownership and responsibility for design decisions.

Research question 3 asked how designers' workflow strategies differ between conditions. The qualitative responses showed two distinct patterns. Manual participants described a direct construction approach, building the layout section by section while continuously comparing their output to the reference. AI-assisted participants described a generate-and-correct approach, in which an initial AI-generated version served as a starting point that was then adjusted through repeated prompting and iterative correction. This distinction reflects the broader shift that Dalsgaard (2017) associates with changes in tool support: as tools take on more generative responsibilities, the designer's role moves away from direct construction and toward evaluation, selection, and refinement.

Across both conditions, participants expressed a preference for a hybrid workflow in which AI supports early structural work while manual refinement handles spacing, typography, alignment, and final detail. This preference points to an important nuance in how AI assistance might be most effectively used. Rather than replacing manual work, AI may be most valuable as a way to establish a structural foundation that designers can then refine with direct control. This also suggests that the binary comparison examined in this study, full AI assistance versus fully manual work, may not reflect the most practical or preferred way of working in real design contexts.

Taken together, the findings suggest that AI assistance in this type of constrained layout task changes how effort, control, and progress are distributed across the design process rather than simply making the task easier. This is consistent with Kittur et al.'s (2013) argument that automation tends to redistribute labour rather than eliminate it, shifting effort toward coordination, verification, and quality control. The study therefore supports the view that AI-assisted design should be evaluated not only in terms of output quality but also in terms of how it shapes the designer's experience of agency and ownership throughout the process.

5.2 Method discussion

The method used in this study had several strengths. The controlled between-subjects design, with both groups working from the same reference layout, starter file, instructions, and time limit, helped reduce variation unrelated to the experimental conditions and supported a focused comparison between the two workflows. The combination of quantitative and qualitative data was also a strength, as it allowed the study to examine both measurable outcomes and the process-level differences that helped explain them. The use of a replication task, rather than a redesign task, was appropriate for the study's focus on precision and structural accuracy, as it provided a common reference point against which both conditions could be assessed.

At the same time, the study has several limitations that should be considered when interpreting the findings. The most significant is the small sample size of five participants per condition. With this sample, the findings are exploratory in nature and should not be taken as broadly generalisable claims about AI-assisted design work. They describe patterns that are meaningful in this specific context but require larger-scale replication before stronger conclusions can be drawn.

A related limitation concerns the fidelity evaluation. The rubric provided a structured basis for comparing layouts, but the scoring was carried out by the researchers themselves without independent verification. Even with predefined criteria, some degree of judgement is unavoidable in deciding how closely a section matches the reference. This introduces the possibility of evaluator bias, and the scores should

therefore be understood as structured assessments rather than fully objective measurements. Future work could strengthen this aspect by involving a second independent evaluator and reporting inter-rater agreement formally.

Screen activity was recorded during the sessions to support qualitative process observation and thematic interpretation of workflow behaviour by the researchers. Due to participant privacy considerations and the exploratory scope of the study, the recordings were not subjected to a separate formal behavioural coding procedure or reported as an independent dataset in the results section. Instead, they were used to support the interpretation of participants' self-reported workflow descriptions and observed interaction patterns during analysis.

The 20-minute time limit was both a methodological strength and a constraint. As a design decision, it standardised time pressure across participants and made sessions manageable. However, the fact that all participants were recorded as time-outs means that completion rate could not function as a differentiating performance measure. The time limit may also have interacted differently with the two workflows: manual participants could continue building directly when time allowed, while AI-assisted participants were sometimes interrupted by waiting for generated outputs. This asymmetry is worth acknowledging when interpreting the performance results.

The decision to treat questionnaire responses and design evaluations as anonymised group-level data, without linking them at the individual participant level, was appropriate for preserving anonymity but limited the analytical possibilities. It was not possible to examine, for example, whether participants who reported lower control also produced lower-scoring layouts, or whether the relationship between workload and performance differed between conditions. Individual-level analysis could have provided a richer understanding of how experience and outcome are connected.

Finally, the study focused specifically on Figma Make as the AI-supported tool. The findings should not be generalised to AI-assisted design tools as a whole, as other tools may offer different interaction models, output quality, and levels of transparency. The results are best understood as specific to this tool, this task, and this participant group, and serve as a basis for further investigation rather than a definitive account of how AI affects UI design work.

6 Conclusions and further research

This chapter presents the main conclusions of the study in relation to the purpose, research questions, and problem statement. It first summarises the key findings and the contribution of the study, and then outlines relevant directions for further research based on the study's limitations and results.

6.1 Conclusions

This study examined how AI-assisted and manual UI design workflows differ in terms of task performance, perceived workload, and workflow strategy in a constrained layout replication task using Figma and Figma Make.

The findings indicate that the two workflows involved a consistent trade-off rather than a simple better-versus-worse outcome. AI assistance supported stronger section-level layout output and was associated with lower perceived workload, suggesting that it can help designers progress further within a fixed time and reduce the burden of the task. At the same time, it did not resolve the overall demands of the task, as all participants were recorded as time-outs regardless of condition. More importantly, the reduction in workload was not accompanied by a stronger sense of agency: manual participants consistently reported higher control over both the process and the result, as well as greater confidence in their submitted layout. This pattern held even though manual work produced lower layout scores under time pressure. The qualitative responses further showed that the two conditions involved structurally different ways of working, direct construction in the manual condition and generate-and-correct in the AI-assisted condition, and that participants across both conditions described a preference for combining the two approaches rather than relying on either alone.

These findings suggest that AI assistance in precision-oriented design tasks changes the balance between efficiency and control rather than improving both simultaneously. A workflow can feel less demanding without making the designer feel more capable or in charge, and output quality can improve without a corresponding increase in perceived agency. This distinction matters for how AI tools are understood and evaluated in design practice: the effects of AI support cannot be assessed by performance or workload alone, because control, confidence, and the structure of the workflow are equally relevant dimensions that may move in the opposite direction to efficiency gains.

6.2 Further research

Several directions for further research follow directly from the findings and limitations of this study. The most immediate priority is replication with a larger sample. With five participants per condition, the present findings are exploratory and cannot support strong claims about effect size or generalisability. A larger study would allow the same patterns to be examined with greater confidence and would make it possible to explore individual-level relationships between workload, control, and performance that could not be analysed here due to the anonymised group-level data structure.

A second direction concerns task type. This study used a constrained replication task specifically to control for variation in creative solutions and to focus the comparison on structural accuracy and workflow. It remains an open question whether the same trade-off between output quality and perceived control would appear in more open-ended tasks, such as concept generation or redesign work, where exploration and creative judgement play a greater role. AI assistance may function differently when there is no fixed reference to work toward.

Third, future research could investigate hybrid workflows more directly. Since participants in both conditions described a preference for combining AI-generated structure with manual refinement, a study that explicitly compares a hybrid condition against fully manual and fully AI-assisted conditions could provide more practically relevant guidance for how AI tools are best integrated into design practice.

Fourth, the relationship between prompt quality and output quality in AI-assisted design work deserves closer attention. Several participants reported difficulty in expressing intended changes clearly enough for the AI system to understand them, and this was identified as a recurring source of friction. Understanding how designers develop prompting strategies over time, and how this affects both performance and perceived control, could inform both tool design and design education.

Finally, future work should examine professional designers alongside student participants. Experience level may significantly shape how the trade-off between efficiency and control is perceived and managed, particularly in tasks that require precision and adherence to predefined requirements. Comparing novice and expert responses to AI assistance could provide insight into whether the costs identified in this study diminish with experience or remain present regardless of skill level.

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8 Appendix

Appendix A

Pre-Study Questionnaire Link

https://prequestionnaire.netlify.app/

Appendix B

Post-Study Questionnaire Link

https://postquestionnaire.netlify.app/

Appendix C

Figma Starter File + Experiment Results

https://www.figma.com/design/hDplL140r93Ug2pVwZT6Pv/Experiment-Results---Instructions?node-id=35-4&t=dC1ykQRCUe8zXOZ0-1

Appendix D

Pre-Task and Post-Task Questionnaire Results

https://docs.google.com/spreadsheets/d/12n6iNa_9HYMiqFNGbX3N5CkERqkUMI2-FrdGRLYQwAQ/edit?usp=sharing

Appendix E

Thesis Section Evaluation with Guidelines

https://docs.google.com/spreadsheets/d/1pJMRxJ-NExvNrCpZlAA8acY3dlPWB0BqJliL3JjlndA/edit?usp=sharing